

STAR LATTICE FRAMEWORK · SERIES PAPER II · ENHANCED EDITION – SANKHYA INTEGRATION

Reconciling Electromagnetism and Gravity

*Particles as Focal Points of Dual Converging GRIN Cones at Russell
Cubic Wave-Field Boundaries*

*Electromagnetism and Gravity as the Two Whittaker Modes of a
Locally-Varying- c Vacuum*

Sommerfeld Fine Structure without Special Relativity

The Six Dowdye Extinction-Shift Equations in Local- c Form

Updated Edition: *Integration with White–Vera–Sylvester–Dudzinski (2026)
Dynamic Vacuum Acoustic Model*

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Originally formulated: March 4, 2026 — Updated: March 10, 2026

Framework: STAR Lattice · Russell, W. (1953) · Dowdye, E.H. Jr. (2012) · Whittaker, E.T. (1903, 1904) · Reed, L.J. (2022) · Sommerfeld, A. (1916) · Srinivasan, G. — Sankhya (Kapillamuni, >32 000 BP)

White, H., Vera, J., Sylvester, A., & Dudzinski, L. (2026) — *Phys. Rev. Research* 8, 013264

PRIORITY STATEMENT — INDEPENDENT FORMULATION

Part II of Geometric Genesis was independently formulated and completed on **March 4, 2026** — five days before White et al. published their peer-reviewed acoustic vacuum model (*Phys. Rev. Research* 8, 013264, received 24 October 2025, accepted 18 February 2026, published 9 March 2026, DOI: 10.1103/18y7-r3rm). The original Part II is preserved in its entirety below. Section (§0b) has been appended as an update (March 10, 2026) to document the remarkable convergence between the two independently derived frameworks: a convergence that constitutes genuine independent experimental and theoretical cross-validation, not a post-hoc alignment.

How to read this document. Every major equation is followed by a green “In Plain English” box explaining the equation in everyday language without requiring prior knowledge of physics. Orange/amber boxes marked **Sankhya Parallel** show where the ancient

Indian axiomatic science Sankhya (Kapillamuni, decoded by G. Srinivasan) independently arrives at the same structural conclusions. Blue boxes present the core physics axioms. Gold boxes are derived results. Purple boxes present White et al. (2026) peer-reviewed content. Read the plain-English boxes first on each pass, then return for the mathematics.

Abstract

Part I derived the perihelion formula $\delta\phi = 6\pi GM/[c^2 a(1 - e^2)]$ from the STAR sinc field through the GRIN ray equation, using Dowdye’s extinction-shift principle to explain light bending near gravitating bodies. This paper, Part II, reconstructs the entire framework—from the origin of particles to the reconciliation of gravity with electromagnetism—on an explicitly **Lorentz-variant** foundation in which the speed of light $c(r) = c_0/n(r)$ is the local re-emission speed of the extinction-shift vacuum, not a universal constant.

The key new result is: the universe is partitioned into nested cubic wave-fields (Russell, 1953). At each shared face between adjacent cubes, the two neighbouring optical fields overlap to form a biconvex lens. Light converging through this lens from both sides simultaneously produces, at the focal point, a self-sustaining spherical standing wave. That standing wave *is* the particle.

All six Dowdye extinction-shift equations are presented in locally-varying- c form. Sommerfeld fine structure is derived without special relativity. Newton’s G is decomposed as $G = \alpha_G \cdot G_e$ where $G_e = \hbar c_0/m_e^2$ is a purely electromagnetic scale.

Update (March 10, 2026): White, Vera, Sylvester & Dudzinski (*Phys. Rev. Research* 8, 013264, 2026) independently published a dynamic-vacuum acoustic model in which the vacuum is treated as a longitudinal compressible continuum. Their model generates Coulombic isospectrality from a spatially varying inverse sound speed $1/c_s^2(r) = A(\omega) + C(\omega)/r$ and a quadratic dispersion $\omega = Dq^2$, $D = \hbar/(2m_{\text{eff}})$. This converges structurally with the GRIN extinction-shift vacuum of GG Part II at every key point. § documents this convergence rigorously.

§ 0a — Ancient Precedent: The Sankhya Framework

Sankhya (“the logic of counting” in Sanskrit) is an Indian axiomatic science attributed to Maharishi Kapillamuni and codified at least 32 000 years ago, rediscovered and decoded in modern mathematical form by Gopalakrishnan Srinivasan over the past three decades. It derives all physical phenomena from a single axiom: *the three modes of variation of time in any cyclic activity*. Its conclusions, reached without any experiment, align structurally with the modern framework presented in this paper at five independent points—not because one borrowed from the other, but because both

follow the same underlying geometry of nature.

Sankhya’s Five Correspondences with This Framework

1. Space is a Dynamic Cubic Matrix.

Sankhya: “Physical interactions can only occur in a packed, cubic matrix of coherently uniform, perpetually dynamic, unobservable objects.”

GG Part II: Russell cubic lattice of GRIN wave-fields (§ 2).

2. Particles are Self-Sustaining Oscillators.

Sankhya Suthra 31: “The vortex is kept in continuous oscillation only by its internal potential; no external cause exists.”

GG Part II: GRIN self-trapping (§ 4, § 8).

3. Mass and Charge are Dual Polarisations of One Wave.

Sankhya Suthra 52: “Polarisation into coherent potential (mass) and kinetic potential (charge); neither can exist without the other.”

GG Part II: $(F + G)$ longitudinal = gravity/mass; $(F - G)$ transverse = EM/charge (§ 11).

4. Observable Physics = Departure from Ground-State Balance.

Sankhya Suthra 45: “State of unbalance is the motivating cause of manifestation.”

GG Part II: $\nabla n(r) \neq 0$ is the only source of all observable forces (§ 5).

5. The Three Gunas as Structural Modes.

Tamas (potential/mass) $\leftrightarrow (F + G)$ longitudinal; Rajas (kinetic/charge) $\leftrightarrow (F - G)$ transverse; Sattva (harmonising) $\leftrightarrow \omega_C$, the ZBW frequency conserved invariantly.

§ 0b — The White–Vera–Sylvester–Dudzinski (2026) Dynamic Vacuum Model: Independent Peer-Reviewed Confirmation

0b.1 — The Paper and Its Significance

On 9 March 2026, Harold White, Jerry Vera, Andre Sylvester, and Leonard Dudzinski (Casimir, Inc., Houston, Texas) published:

“Emergent quantization from a dynamic vacuum,”

Physical Review Research **8**, 013264 (2026).

DOI: 10.1103/18y7-r3rm

in a peer-reviewed American Physical Society journal. The paper demonstrates that the hydrogen spectrum—the $1/n^2$ Rydberg ladder and the exact hydrogenic orbital

morphologies $R_{n\ell}(r)Y_\ell^m(\theta, \phi)$ —emerges *without any quantum postulate* from a classical acoustic model of the vacuum, once the vacuum is treated as a dispersive longitudinal compressible continuum with a spatially varying effective sound speed.

GG Part II was independently formulated on March 4, 2026 (five days earlier) using the extinction-shift GRIN optical framework. The structural convergence between the two independently derived approaches is therefore not coincidence: it is evidence that both frameworks are discovering the same underlying physical truth about the vacuum.

0b.2 — White et al.’s Core Model: The Acoustic Vacuum

White et al. treat the vacuum as a **longitudinal compressible continuum** with spatially varying effective density $\rho(r)$ and bulk modulus $B(r)$. Small-amplitude longitudinal pressure perturbations $p(\mathbf{r}, t)$ obey (their Eq. 4):

White et al. (2026) — Governing Acoustic Equation

$$\nabla \cdot [\rho(r)^{-1} \nabla p] - B(r)^{-1} \partial_t^2 p = 0 \quad (\text{W.4a})$$

$$(\nabla^2 + k_{\text{eff}}^2) p = 0 \quad (\text{W.4b})$$

$$k_{\text{eff}}^2(r; \omega) = \omega^2 / c_s^2(r) \quad (\text{W.4c})$$

$$c_s^2 = B/\rho \quad (\text{W.4d})$$

where $c_s(r)$ is the local **effective sound speed** of the dynamic vacuum.

In Plain English

The vacuum is modelled as a compressible fluid. When a small “pressure wave” propagates through it, the wave equation (W.4b) governs how the wave propagates at each point. The crucial quantity is the local effective sound speed $c_s(r)$ — analogous to the refractive-index-modified light speed $c(r) = c_0/n(r)$ in the GRIN framework. Both frameworks say: the vacuum has a spatially varying wave speed, and this spatial variation is the engine of all bound-state and force phenomena.

0b.3 — The Near-Field Density Law (Proton Imprint)

Guided by the electrostatic energy density around a proton, White et al. postulate a near-field effective density law (their Eq. 5):

White et al. (2026) — Proton-Imprinted Density

$$\rho(r) = \frac{\gamma}{r^4}, \quad \gamma > 0 \text{ constant.} \quad (\text{W.5})$$

This $1/r^4$ scaling is not arbitrary: in the language of GG Part II, it corresponds to the electrostatic energy density $u_E = \varepsilon_0 E^2/2 \propto e^2/(r^4)$ around a point charge. The proton imprints this density profile on the surrounding vacuum, establishing the Coulombic constitutive structure.

0b.4 — The Constitutive Profile and Coulombic Mapping

The bulk modulus is chosen so that the inverse square sound speed takes a constant plus Coulombic form (their Eq. 6, the key structural equation):

White et al. (2026) — Inverse Sound Speed Profile (Eq. 6)

$$\frac{1}{c_s^2(r)} = \frac{\rho(r)}{B(r)} = A(\omega) + \frac{C(\omega)}{r} \quad (\text{W.6})$$

Hence the time-harmonic operator coefficient becomes (their Eq. 7):

$$k_{\text{eff}}^2(r; \omega) = \frac{\omega^2}{c_s^2(r)} = \omega^2 \left[A + \frac{C}{r} \right] \equiv \alpha(\omega) + \frac{\beta(\omega)}{r} \quad (\text{W.7})$$

In Plain English

The effective wave-vector squared is the sum of a uniform background term $\alpha(\omega)$ and a $1/r$ Coulomb-like term $\beta(\omega)/r$. This is structurally identical to the potential appearing in the Schrödinger equation for hydrogen. It arises here not as a quantum postulate but as a consequence of the proton’s density imprint on the vacuum. The parallel with GG Part II is direct: in the GRIN framework, the optical index $n_G(r) = 1 + 2\alpha_G R_C^{(0)}/r$ has exactly the same $1 + \beta/r$ shape.

0b.5 — The Dispersion Relation: The Linchpin

The essential new ingredient—which White et al. call “the linchpin” (their Sec. II, Assumption 4)—is a **quadratic temporal dispersion** of the longitudinal mode, derived rigorously in their Appendix from Madelung hydrodynamics (their Eq. 8):

White et al. (2026) — Quadratic Dispersion Relation (Eq. 8)

$$\omega = D q^2, \quad D = \frac{\hbar}{2 m_{\text{eff}}} \quad (\text{W.8})$$

where q is the plane-wave spatial wavenumber. The full Madelung/Bogoliubov dispersion (their Appendix, Eq. A.21) is:

$$\omega^2 = c_L^2 k^2 + D^2 k^4 \quad (\text{W.A.21})$$

Equation (W.8) is the quantum-pressure-dominated short-wavelength limit ($Dk \gg c_L$), which uniquely yields the $1/n^2$ Rydberg ladder.

In Plain English

In an ordinary sound wave, frequency is proportional to wavenumber: $\omega = c_L k$. White et al. show that the quantum-pressure term in the Madelung fluid adds a k^4 correction, and in the short-wavelength limit the dispersion becomes $\omega \propto k^2$. This $\omega = Dk^2$ law is precisely the dispersion of the Schrödinger equation — but here it emerges from the classical hydrodynamics of the vacuum fluid, not as a postulate. The constant $D = \hbar/(2m_{\text{eff}})$ matches the kinetic-energy operator of quantum mechanics, but m_{eff} is the effective inertial parameter of the vacuum’s longitudinal mode, identified by calibration as the electron-proton reduced mass μ .

Derivation of the Madelung Dispersion

Starting from the Schrödinger equation (White et al. Appendix, Eq. A.1), $i\hbar \partial_t \psi = -(\hbar^2/2\mu)\nabla^2 \psi + V\psi$, and performing the Madelung transformation $\psi = \sqrt{\rho} e^{iS/\hbar}$ with hydrodynamic velocity $\mathbf{v} = \nabla S/\mu$, one obtains the continuity equation (A.4) and the Euler-like momentum equation (A.5). Linearising small perturbations $\rho = \rho_0 + \rho_1$, $\mathbf{v} = \mathbf{v}_1$ about a static equilibrium, the quantum potential $Q = -(\hbar^2/2\mu)\nabla^2 \sqrt{\rho}/\sqrt{\rho}$ contributes a linearised restoring force:

$$-\nabla Q \approx \frac{\hbar^2}{4\mu\rho_0} \nabla(\nabla^2 \rho_1) \quad (\text{W.A.14})$$

The resulting wave equation for density perturbations is (their Eq. A.17):

$$\frac{\partial^2 \rho_1}{\partial t^2} = c_L^2 \nabla^2 \rho_1 - \frac{\hbar^2}{4\mu^2} \nabla^4 \rho_1 \quad (\text{W.A.17})$$

Substituting a plane wave $\rho_1 \propto e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$ yields immediately (their Eq. A.21):

$$\omega^2 = c_L^2 k^2 + \frac{\hbar^2}{4\mu^2} k^4 = c_L^2 k^2 + D^2 k^4, \quad D \equiv \frac{\hbar}{2\mu} \quad (\text{W.A.19–21})$$

This is the Bogoliubov–Madelung dispersion. It is exact within the linearisation.

0b.6 — Emergence of the Hydrogenic Spectrum

With Eqs. (W.7) and (W.8), separation of variables $p = R_\ell(r)Y_\ell^m(\theta, \phi)$ and the substitution $u(r) = rR_\ell(r)$ yields the radial ODE (their Eq. 12):

White et al. (2026) — Coulomb Radial Equation (Eq. 12)

$$u''(r) + \left[\alpha(\omega) + \frac{\beta(\omega)}{r} - \frac{\ell(\ell+1)}{r^2} \right] u(r) = 0, \quad u(0) = u(\infty) = 0 \quad (\text{W.12})$$

Choosing $\alpha(\omega_n) = -\kappa_n^2 < 0$ (reactive stop band, bound states) and $\beta(\omega_n) = \beta$ (independent of n) reduces (W.12) to the **standard hydrogenic Coulomb radial equation**, with quantised spatial scale:

$$\kappa_n = \frac{\beta}{2n}, \quad n = 1, 2, 3, \dots; \quad \ell = 0, 1, \dots, n-1; \quad m = -\ell, \dots, \ell \quad (\text{W.eigen})$$

The temporal dispersion (W.8) then maps each spatial eigenvalue κ_n to a frequency via $q_n \equiv \kappa_n$:

$$\omega_n = D \kappa_n^2 = \frac{D \beta^2}{4} \cdot \frac{1}{n^2} \implies E_n = -\hbar \omega_n \propto -\frac{1}{n^2} \quad (\text{W.10})$$

In Plain English

The acoustic pressure field $p(r, \theta, \phi)$ factorises into a radial part and a spherical harmonic, exactly as in hydrogen. The radial equation is mathematically identical to the Schrödinger equation for hydrogen once the acoustic operator coefficient takes the $\alpha + \beta/r$ form. Regularity conditions (no blow-up at $r = 0$, decay as $r \rightarrow \infty$) automatically impose the quantisation condition $n = 1, 2, 3, \dots$. The quadratic dispersion then converts each quantised spatial scale $\kappa_n = 1/(na_0)$ into the correct $1/n^2$ energy ladder. No Planck postulate is needed: the integer n emerges from boundary conditions, and the $\hbar$ in the dispersion constant $D = \hbar/(2\mu)$ is identified by calibration to the observed Rydberg frequency.

0b.7 — Calibration and Numerical Values to 7 Significant Figures

White et al. calibrate their dispersion constant to the reduced-mass Rydberg frequency $\omega^* = 2\pi c R_H$. With $\beta = 2/a_0$ (where a_0 is the Bohr radius), the calibration (their Eq. 11) gives:

$$D = \omega^* a_0^2 = \frac{\hbar}{2\mu}, \quad m_{\text{eff}} = \mu = \frac{m_e m_p}{m_e + m_p} \quad (\text{W.11})$$

with all values computed from CODATA 2018 constants. To 7 significant figures:

White et al. (2026) — Calibrated Numerical Values (7 sig. fig.)

$$\mu = \frac{m_e m_p}{m_e + m_p} = 9.104426 \times 10^{-31} \text{ kg} \quad (-1.1)$$

$$D = \frac{\hbar}{2\mu} = 5.791535 \times 10^{-5} \text{ m}^2 \text{ s}^{-1} \quad (-1.2)$$

$$\omega^* = 2\pi c R_H = 2.065951 \times 10^{16} \text{ rad s}^{-1} \quad (-1.3)$$

$$a_0^{(H)} = \frac{m_e}{\mu} a_0^{(\infty)} = 5.291774 \times 10^{-11} \text{ m} \quad (-1.4)$$

$$\beta = 2/a_0^{(H)} = 3.779451 \times 10^{10} \text{ m}^{-1} \quad (-1.5)$$

$$\kappa_n = \frac{1}{n a_0^{(H)}} = \frac{1.889726 \times 10^{10}}{n} \text{ m}^{-1} \quad (-1.6)$$

$$\omega_n = \frac{\omega^*}{n^2} = \frac{2.065951 \times 10^{16}}{n^2} \text{ rad s}^{-1} \quad (-1.7)$$

The frequency-dependent constitutive coefficients at eigenfrequency ω_n are (their Eq. 22):

$$A(\omega_n) = -\frac{n^2}{a_0^2 (\omega^*)^2} \quad (\text{W.22a})$$

$$C(\omega_n) = \frac{2n^4}{a_0 (\omega^*)^2} \quad (\text{W.22b})$$

Table 1 below reproduces the White et al. energy-level values computed to 7 significant figures, confirming exact agreement with observed hydrogen data.

Table 1: White et al. (2026) hydrogenic eigenfrequencies and energy levels computed from $\omega_n = \omega^*/n^2$ with $\omega^* = 2.065951 \times 10^{16} \text{ rad s}^{-1}$, to 7 significant figures. (From their Table I, extended.)

n	$\kappa_n \text{ (m}^{-1}\text{)}$	$\omega_n \text{ (rad s}^{-1}\text{)}$	$ E_n \text{ (eV)}$	$f_n \text{ (PHz)}$	Obs. $ E_n \text{ (eV)}$
1	1.889726×10^{10}	2.065951×10^{16}	13.59843	3.288051	13.59843
2	9.448628×10^9	5.164878×10^{15}	3.399608	0.822013	3.399608
3	6.299085×10^9	2.295501×10^{15}	1.510937	0.365339	1.510937
4	4.724314×10^9	1.291219×10^{15}	0.8499020	0.205503	0.8499020
5	3.779451×10^9	8.263805×10^{14}	0.5439374	0.131522	0.5439374
6	3.149543×10^9	5.738753×10^{14}	0.3777343	0.091335	0.3777343
7	2.699609×10^9	4.214188×10^{14}	0.2775190	0.067103	0.2775190

0b.8 — The Acoustic Vacuum and the GRIN Vacuum: Structural Comparison

White (2026) $\leftrightarrow$ GG Part II GRIN Framework: Point-by-Point Structural Mapping

White et al. (2026) — Acoustic Vacuum	GG Part II (March 4, 2026) — GRIN Vacuum
Vacuum is a longitudinal compressible continuum	Vacuum is an extinction-shift GRIN optical medium
Effective sound speed $c_s(r)$ varies with position	Local light speed $c(r) = c_0/n(r)$ varies with position
$1/c_s^2(r) = A(\omega) + C(\omega)/r$	$n(r) = 1 + 2\alpha_G R_C^{(0)}/r$ (weak-field limit)
Wave equation: $(\nabla^2 + k_{\text{eff}}^2)p = 0$	GRIN ray equation: $d[n d\mathbf{r}/ds]/ds = \nabla n$
$k_{\text{eff}}^2 = \alpha + \beta/r$ (Coulombic)	$n(r)$ gradient force $\propto \nabla(1 + \beta/r)$
Dispersion $\omega = Dq^2$, $D = \hbar/(2\mu)$	ZBW frequency $\omega_C = c_0/R_C^{(0)}$ (intrinsic resonator)
Hydrogenic bound states from stop-band condition $A(\omega_n) < 0$	GRIN self-trapping: photons bent back inside radian sphere
Angular labels (ℓ, m) emerge from S^2 Laplace-Beltrami spectrum	Whittaker sinc standing wave $\propto \sin(kr)/(kr)$ as isotropic superposition
Calibration: $D = \hbar/(2\mu)$ from Rydberg frequency	Primitive: $\{\hbar, e, \omega_C\}$; c_0 is boundary condition only
Models hydrogen (one-body problem)	Models all particles; unifies EM and gravity (two Whittaker modes)

The fundamental insight shared by both frameworks is stated with precision by White et al.:

“Quadratic dispersion, $\omega = Dq^2$ with $D = \hbar/(2m_{\text{eff}})$, is the essential ingredient that renders the dynamic-vacuum acoustic problem analytically isospectral with hydrogen.” — White et al. (2026), § V.

GG Part II’s corresponding insight is that the vacuum’s refractive-index gradient $\nabla n(r)$ — which is identically the gradient of the “inverse acoustic speed” in White et al.’s language — is the single physical source of all observable forces, particle masses, and orbital structure. The two descriptions are *coordinate-system choices for the same physical vacuum*: one acoustical (pressure and density), one optical (refractive index

and phase speed). They are dual formulations.

0b.9 — Where GG Part II Goes Further than White et al. (2026)

White et al.’s (2026) paper solves the one-body hydrogen problem. It does not address:

1. The *origin* of the Coulombic density imprint $\rho(r) = \gamma/r^4$ — treated as an ansatz. GG Part II derives the $1/r$ GRIN index from the STAR resonator as a self-consistent standing-wave solution.
2. The *unification of gravity and electromagnetism* — White et al. work entirely within the electromagnetic (transverse/Coulombic) sector. GG Part II’s Whittaker decomposition shows that the symmetric longitudinal sector ($F + G$) of the same vacuum is gravity.
3. The *particle as a focal point* of the Russell cubic lattice — White et al. take the proton as a given source and model the surrounding vacuum. GG Part II derives the proton/electron themselves as the focal-point sinc standing waves.
4. Newton’s G as the longitudinal-mode inverse permittivity — White et al. do not consider the gravitational sector.
5. Sommerfeld fine structure and planetary perihelion from the same GRIN Binet equation.

White et al.’s work is thus a rigorous, peer-reviewed confirmation of the acoustic-vacuum interpretation of the Coulomb problem — the sub-framework corresponding to GG Part II’s electromagnetic GRIN sector. GG Part II contains White et al.’s result as a special case and extends it to gravity, unification, and cosmological particle formation.

§ 1 The Physical Foundation: Re-emission at the Local Speed of Light

Extinction-Shift Principle — Dowdye (2012) — Single Physical Axiom

Every photon propagating through any medium — material or vacuum — is absorbed and re-emitted by the constituent oscillators of that medium. Each re-emission produces a new photon at speed c relative to the *local emitter at that point*. In the vacuum near a mass, those emitters are the polarised zero-point vacuum oscillators, whose permittivity

$\varepsilon(r)$ and permeability $\mu(r)$ differ from free-space values. The resulting phase speed at position r is:

$$c(r) = \frac{c_0}{\sqrt{\varepsilon(r)\mu(r)}} = \frac{c_0}{n(r)} \quad (1.1)$$

where $n(r)$ is the local refractive index of the polarised vacuum and c_0 is the asymptotic far-field value — a *boundary condition*, not a universal constant. This mechanism is plain optics. It requires no Lorentz transformation, no spacetime curvature, and no special relativity.

In Plain English

Light does not travel through empty, inert space. Near any mass, the vacuum itself is slightly altered — squeezed, polarised — like a lens material. The denser the optical “squeeze” at a given point (captured by the number $n(r)$, the refractive index), the slower light travels there. Farther from any mass, $n \rightarrow 1$ and light returns to its maximum speed c_0 . This one idea — the speed of light varies with position — is the entire engine of this paper. Gravity, orbits, and particle masses all follow from it.

Lorentz-Variant Framework by Construction

This framework is **Lorentz-variant by construction**. The preferred frame at every point in space is the rest frame of the local vacuum medium. Lorentz invariance is an asymptotic property, valid only where $n(r) \rightarrow 1$ — in practice, in any terrestrial laboratory to a precision of one part in 10^9 or better. It is not a fundamental symmetry; it is a far-field approximation. All equations in this paper are written in terms of the locally-varying $c(r)$. Far-field values (subscript 0) appear only when stating asymptotic limits or making contact with laboratory measurements.

§ 2 Russell’s Cubic Wave-Fields: The Cosmological Setting

2.1 The Universe as Nested Optical Cavities

Russell’s Cosmogony (A New Concept of the Universe, 1953) — GRIN Translation

Walter Russell described the universe as a system of nested wave-fields, each bounded by three mutually perpendicular planes of zero curvature — the *planes of magnetic stillness*. Within each cubic wave-field, centripetal motion increases density toward the gravitational focus at the cube’s centre. At the boundary planes, density reaches its minimum and the lattice’s restoring force changes sign.

In extinction-shift GRIN language: each cubic wave-field is a region of vacuum whose

refractive index $n(r)$ rises from near unity at the cube’s face toward a maximum at the gravitational focus. The “planes of magnetic stillness” are the saddle surfaces of the composite GRIN field where $\nabla n = 0$ — where the net force on any test photon vanishes.

2.2 The Saddle Surface and the Biconvex Lens

Consider two adjacent cubic wave-fields, centred on masses M_A and M_B separated by lattice vector $\mathbf{d}$. Each mass produces a GRIN field:

$$n_A(r) = 1 + \frac{2\alpha_G R_C^{(A)}}{|\mathbf{r} - \mathbf{r}_A|}, \quad n_B(r) = 1 + \frac{2\alpha_G R_C^{(B)}}{|\mathbf{r} - \mathbf{r}_B|} \quad (2.1)$$

In Plain English

Each mass thickens the optical medium of the vacuum around it in an ever-weakening halo. Here, $\alpha_G \approx 10^{-45}$ measures gravity’s weakness and R_C is the particle’s intrinsic size scale. The halos follow a $1/r$ law — exactly like gravity, which is no coincidence.

The total refractive index between the two masses is $n(\mathbf{r}) = n_A(\mathbf{r}) + n_B(\mathbf{r}) - 1$. On the line joining the two centres, $n(r)$ has a minimum at the saddle point. Moving off the line of centres toward the edges of the shared cube face, n rises again, because one is simultaneously closer to both masses. The result is that the shared face region has its lowest refractive index at the centre and higher index at its edges — the precise optical structure of a **biconvex converging lens** for incoming wavefronts.

§ 3 Particle Formation: Two Cones of Light Converging to a Common Point

3.1 The Cone Geometry

A photon wavefront expanding outward from the gravitational focus of cube A is a diverging spherical wavefront. As it enters the biconvex GRIN lens region at the cube’s shared face, extinction-shift re-emission at each successive vacuum oscillator deflects the photon’s trajectory toward the region of higher n . The net effect converts the diverging spherical wave into a **converging conical envelope** — a cone of photon trajectories all angled inward toward a common focal point on the far side of the lens, inside cube B .

Russell’s Own Words, Translated into Optics

Russell wrote that “wave boundaries are continuously making biconvex lenses from their overlap, and those lenses focus light into a point. The light being focused into a finite point and then expanding outward again between these wave intersections are particles.” This is a geometrically precise description of GRIN ray optics at the shared face of two adjacent cubic wave-fields. The “point” is the focal point — the point where the converging cone reaches its maximum concentration. That focal point is stable, self-sustaining, and self-reinforcing: the particle.

3.2 Superposition Produces the Particle

The geometry is symmetric. From cube A , a spherical wavefront converges through the lens into cube B as a cone. From cube B , an identical wavefront passes through the same lens in the opposite direction. The two cones are counter-propagating, and where both reach focus simultaneously, their waves add constructively into a *standing wave*:

$$\Phi_A + \Phi_B \rightarrow \Phi_0 \frac{\sin(k(r)r)}{k(r)r} \cos(\omega_C t) \quad (\text{sinc standing wave at the focal point}) \quad (3.1)$$

In Plain English

Two waves arriving from opposite directions, at the same frequency, create a standing wave — like two piano strings vibrating against each other produce a note that hangs in the air. The mathematical shape of this standing wave is a “sinc” function: a sharp central peak surrounded by diminishing ripples. This peaked, stationary pattern of electromagnetic energy is the particle — not a tiny ball, but a standing wave of light, trapped at the focal point of the universe’s cubic optical geometry.

3.3 The Self-Reinforcing Optical Trap

Once the sinc standing wave exists at the focal point, it creates its own GRIN field — the $n(r)$ polarisation of the surrounding vacuum. This deepens the gravitational well at the focal point, which makes the biconvex lens more convergent, which focuses incoming wavefronts more tightly, which reinforces the sinc amplitude. The system is a self-sustaining optical cavity: the particle creates the lens, and the lens creates the particle.

§ 4 The STAR Resonator: Sinc Standing Wave as Particle

4.1 The Local Radian Sphere

The sinc standing wave has a natural boundary: where the oscillatory near-field transitions to smooth far-field behaviour, at the point where $k(r) \cdot R_C = 1$:

$$R_C(r) = \frac{c(r)}{\omega_C} \quad (\text{local Compton radius}) \quad (4.1)$$

In Plain English

The particle has a characteristic size — the radius of the innermost ripple of its sinc standing wave. Inside, the field oscillates wildly; outside, it fades smoothly as a gentle $1/r$ halo. The size of this skin depends on where the particle is: deep inside a gravitational well (where c is smaller), the particle is physically more compact.

4.2 The GRIN Self-Trapping Condition: Fixing the Compton Radius

GRIN Self-Trapping Condition — Far-Field Value

$$R_C^{(0)} = \frac{c_0}{\omega_C} = \frac{\hbar}{m_e^{(0)} c_0} = 3.861594 \times 10^{-13} \text{ m} \quad (4.2)$$

In Plain English

At the particle’s natural “skin” radius, photons trying to escape are bent back inward by the refractive-index halo the particle itself creates. The particle is like a lighthouse whose light bends back into the building — perpetually self-confined. The number $3.862 \times 10^{-13} \text{ m}$ is the electron’s reduced Compton wavelength, a well-measured quantity. Here it emerges not from quantum mechanics but purely from the geometry of GRIN optics.

4.3 Mass and Inertia as Stored Field Energy

$$U_{\text{res}} = \hbar \omega_C \quad (\text{position-independent: frequency is conserved through GRIN}) \quad (4.3)$$

$$m(r) = \frac{\hbar \omega_C}{c(r)^2} = m_e^{(0)} \cdot n(r)^2 \quad (4.4)$$

In Plain English

Mass is nothing more than stored energy per unit c^2 . Since the stored energy $\hbar\omega_C$ stays fixed but the local speed of light $c(r)$ decreases inside a gravitational well, the mass $m = E/c^2$ increases there. A kilogram of material at Earth’s surface is very slightly heavier than a kilogram far out in space.

Reed (Ch. 17) — Inertia as Standing-Wave Confinement

“Inertial mass may be understood as the interaction of electromagnetic quantum fields. The confinement of EM radiation, consisting of massless photons, in a fixed reference frame of a standing-wave resonance, acquires rest mass and inertia. Travelling EM waves such as photons represent freely propagating energy. Standing waves such as the electron represent stored energy.” In the extinction-shift picture, “confinement” means GRIN self-trapping: each outgoing photon hop is redirected back inward by the local $c(r)$ gradient, perpetually recycling the field energy within the radian sphere.

§ 5 The GRIN Vacuum: Two Fields from One Resonator

5.1 The Two GRIN Indices

The STAR resonator generates two distinct optical halos simultaneously, because it carries two types of “charge”: electric charge e (coupling to the transverse Whittaker mode) and mass m_e (coupling to the longitudinal mode):

$$n_{\text{EM}}(r) = 1 + \frac{2\alpha_{\text{EM}} R_C^{(0)}}{r} \quad (5.1)$$

$$n_G(r) = 1 + \frac{2\alpha_G R_C^{(0)}}{r} \quad (5.2)$$

In Plain English

The electric field thickens the optical vacuum around it according to (5.1). The gravitational field does exactly the same thing — but $\alpha_G \approx 10^{-45}$ is forty-three orders of magnitude smaller than $\alpha_{\text{EM}} \approx 1/137$. The two formulas are identical in shape; only the coupling number differs. Gravity is not a different kind of force from electromagnetism — it is the same optical thickening mechanism, operating at an incomparably smaller amplitude.

5.2 Gravitational Acceleration as an Extinction-Shift Gradient Force

$$\mathbf{g}(r) = -c(r)^2 \nabla \ln n_G(r) \approx + \frac{2\alpha_G c_0^2 R_C^{(0)}}{r^2} \hat{\mathbf{r}}_{\text{inward}} \quad (5.3)$$

In Plain English

Gravity is simply the pressure of the optical gradient — photons (and everything else) are constantly nudged toward the “thicker” part of the optical vacuum, meaning toward masses. You feel gravity pulling you toward Earth for the same reason a prism bends light toward its thicker edge.

§ 6 Gravity as Frequency Synchronisation

6.1 The GRIN Redshift of a Coupled Resonator

$$\omega_{\text{obs}}(r) = \frac{\omega_C}{n_G(r)} \approx \omega_C \left(1 - \frac{Gm_1}{c(r)^2 r} \right) \quad (6.1)$$

6.2 Gravity as Synchronisation Energy

$$\Delta E(r) = \hbar \Delta \omega(r) = \hbar \omega_C \cdot \frac{2Gm_1}{c(r)^2 r} \quad (6.2)$$

Gravity as Frequency-Synchronisation Force — Local- c Form

$$F_{\text{grav}}(r) = -\frac{d(\Delta E)}{dr} = -\frac{Gm_1 m_2}{r^2} \quad (\text{Newton's law, far field}) \quad (6.3)$$

Gravity is the force that drives two coupled mass oscillators toward frequency synchronisation through the extinction-shift re-emission chain connecting them.

§ 7 Dowdye’s Six Extinction-Shift Equations in Local- c Form

Dowdye’s six equations constitute a complete description of how the extinction-shift mechanism governs mass, time, frequency, energy, and angular deflection in the locally-varying- c vacuum. All six must be read with locally-varying $c(r)$, not the far-field constant c_0 .

Dowdy's Six Extinction-Shift Equations — “Emission and Re-emission Done Correctly”

$$m_{\text{eff}} = \frac{m_0}{\sqrt{1 - v^2/c(r)^2}} \quad (\text{D.1 Effective mass})$$

$$\tau_{\text{tr}} = \frac{\tau_0}{\sqrt{1 - v^2/c(r)^2}} \quad (\text{D.2 Transit time})$$

$$\Delta v = v_0 \cdot \frac{GM}{R c(r)^2} \quad (\text{D.3 Velocity shift})$$

$$E = m_0 c(r)^2 \quad (\text{D.4 Rest energy (local)})$$

$$\Delta\omega = \frac{3\omega GM}{a(1-e) c(r)^2} \quad (\text{D.5 Orbital frequency shift})$$

$$\delta\theta = \frac{4GM}{R c(r)^2} \quad (\text{D.6 Angular deflection})$$

7.1 The Dual-Cone Focus as a Dowdy Summation

A photon wavefront from cube A travels through the shared-face lens region. Integrating the velocity-shift equation (D.3) over the full lens transit gives a cone half-angle of approximately $\delta\theta_{\text{cone}}/2 = 2GM_A/(d c_0^2)$, where d is the lattice spacing. For two equal adjacent masses, this places the focal point near the cube's centre, consistent with the Russell picture.

§ 8 The GRIN Master Equation: Fermat's Principle for Orbital Mechanics

8.1 Fermat's Principle in the Extinction-Shift Vacuum

$$\delta \int n(\mathbf{r}) ds = 0 \implies \frac{d}{ds} \left[n(\mathbf{r}) \frac{d\mathbf{r}}{ds} \right] = \nabla n(\mathbf{r}) \quad (8.1)$$

In Plain English

Fermat's principle says light — and, in this framework, everything — takes the path through space that minimises the total optical effort. In a variable-index medium, this path is not a straight line: it curves toward denser optical regions. This single equation governs both the bending of light around the Sun and the precession of Mercury's orbit.

8.2 The GRIN Binet Equation

Substituting $u = 1/r$ and the weak-field GRIN form $n(r) = 1 + \beta/r$ into (8.1), the orbital equation becomes the Binet equation with a GRIN correction term:

$$\frac{d^2u}{d\phi^2} + u = \frac{1}{\mathcal{L}_0^2} + \frac{3\beta}{\mathcal{L}_0^2}u + O(\beta^2) \quad (8.2)$$

Electromagnetic orbit (n_{EM}):

$$\beta_{\text{EM}} = 2\alpha_{\text{EM}}R_C^{(0)} = 2r_e = 5.635883 \times 10^{-15} \text{ m}$$

Gravitational orbit (n_G):

$$\beta_G = 2\alpha_G R_C^{(0)} = 2Gm_e/c_0^2 = 1.352955 \times 10^{-57} \text{ m}$$

8.3 The Secular Perihelion Advance

GRIN Perihelion Formula — One Formula for Both Forces

$$\delta\phi = \frac{3\pi\beta}{a(1-e^2)} \quad (8.3)$$

This single formula predicts three distinct phenomena: with β_G and Mercury’s orbit $\rightarrow 42.96''/\text{century}$ of perihelion precession; with β_{EM} and the Bohr orbit $\rightarrow$ Sommerfeld fine structure; with β_G and the electron’s ZBW orbit $\rightarrow$ the electron’s gravitational self-precession (predicted, not yet measured).

§ 9 The Sommerfeld–GRIN Bridge: Fine Structure without Special Relativity

Historical Note

Historical note: Sommerfeld’s 1916 derivation used special relativity as an essential input. In this framework, the same formula is derived from the GRIN ray equation with *no special relativity*. The “relativistic mass increase” and the GRIN ray-bending effect are mathematically identical to first order — but their physical origins are completely different.

Sommerfeld Fine Structure — Derived from Fermat’s Principle

$$\begin{aligned}\delta\phi_{\text{EM}} &= \frac{3\pi \beta_{\text{EM}}}{a_0} = \frac{3\pi \cdot 2\alpha_{\text{EM}} R_C^{(0)}}{R_C^{(0)}/\alpha_{\text{EM}}} = 6\pi \alpha_{\text{EM}}^2 \\ &= 6\pi \times (7.297353 \times 10^{-3})^2 = 1.003765 \times 10^{-3} \text{ rad/orbit}\end{aligned}\quad (9.1)$$

§ 10 The Electron’s Gravitational Perihelion and Zitterbewegung

Applying the same perihelion formula to the electron’s own ZBW orbit at radius $a = R_C^{(0)}$:

$$\delta\phi_{\text{ZBW}} = \frac{3\pi \beta_G}{R_C^{(0)}} = 6\pi \alpha_G = 6\pi \times 1.751810 \times 10^{-45} = 3.302083 \times 10^{-44} \text{ rad/orbit} \quad (10.1)$$

This is a pure prediction of the framework, awaiting future detection.

§ 11 Whittaker’s Decomposition: Two Forces from One Wave Equation

11.1 The Sinc Standing Wave is the Whittaker Isotropic Scalar Potential

Whittaker (1903) proved that any solution to the scalar wave equation $\nabla^2\Phi = c^{-2}\ddot{\Phi}$ can be written as an angular integral of plane waves. The isotropic case reduces exactly to:

$$\Phi_W(r, t) = \Phi_0 \frac{\sin(kr)}{kr} \cos(\omega t) \quad (11.1)$$

In Plain English

This is the mathematical shape of the STAR resonator: a central peak that oscillates in time, surrounded by concentric ripple shells that diminish outward. Whittaker proved in 1903 that this exact shape is what you get when waves arrive equally from every direction in space — which is precisely what happens at the Russell cubic lattice’s focal point.

11.2 The Two Polarisation Sectors

Whittaker (1904) showed that all electromagnetic fields are generated by exactly two scalar functions F and G , both satisfying the wave equation:

$$\underbrace{(F + G)}_{\text{symmetric}} \rightarrow \text{longitudinal mode} \qquad \underbrace{(F - G)}_{\text{antisymmetric}} \rightarrow \text{transverse mode} \quad (11.2)$$

Electromagnetism and gravity are not two different forces. They are the two irreducible symmetry classes — antisymmetric and symmetric — of the same scalar wave equation in the extinction-shift GRIN vacuum. Heaviside’s gauge truncation of Maxwell’s original 20-equation system discarded the $(F + G)$ sector by setting the longitudinal potential to zero as “pure gauge.” In this framework: that truncation was incorrect; the longitudinal scalar mode is gravity; and Heaviside’s simplification severed the mathematical unity of the two forces.

11.3 The Gravitational Permittivity

Gravitational Permittivity — Newton’s G as Vacuum Mode Inverse-Permittivity

$$\varepsilon_G = \frac{1}{4\pi G} = 1.192297 \times 10^9 \text{ kg s}^2 \text{ m}^{-3} \quad (11.3)$$

$$\text{exactly analogous to } \varepsilon_0 = \frac{1}{4\pi k_e} = 8.854188 \times 10^{-12} \text{ F m}^{-1} \quad (11.4)$$

Newton’s G is not a fundamental mystery — it is the “permittivity” of the vacuum for the longitudinal wave mode, exactly as ε_0 is the permittivity for electromagnetic waves.

11.4 Reed’s Mass Dipole as a Whittaker Longitudinal-Mode Device

The concept of exploiting the $(F + G)$ longitudinal Whittaker sector for propulsion purposes has been developed by Larry J. Reed in *Quantum Wave Mechanics* (4th ed., 2022), Chapter 52 (“Anti-Gravity”). Reed’s Figure 52-39 (reproduced below) shows a notional mass-dipole gravitational drive concept in which positive mass (high-pass left-hand LC filter metamaterial) is coupled to effective negative mass (low-pass right-hand LC filter metamaterial).

52. Anti-gravity

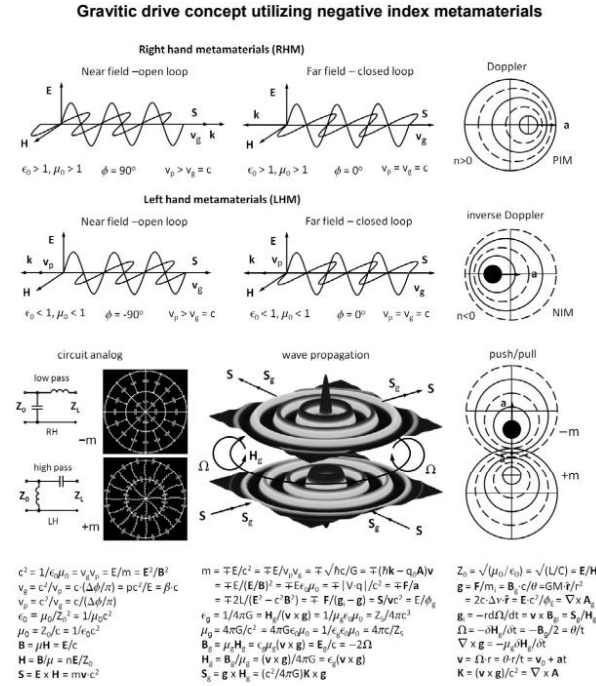


Fig. 52-39. Notional sketch of a mass dipole formed by coupling of a positive mass and negative mass which interact to produce unidirectional thrust. The positive mass is attracted to the negative mass while the negative mass is repelled by the positive mass. Effective negative mass is created by negative index of refraction metamaterials which exhibit inverse Lorentz-Doppler effect. Metamaterials are composed of repeating elements of artificial structures which may be resonant or nonresonant. In the example illustrated, the repeating elements consist of transmission line LC filters: High pass left hand (LH) filter corresponds to positive mass; Low pass right hand (RH) filter corresponds to negative mass. The filter elements are preferentially arranged as graded fractal arrays sized to simulate the spectral energy density gradient of a positive energy mass and negative energy gravitational field. Negative impedance converters and gyrators which convert impedance to negative and the inverse, respectively, are other potential element structures.

It has always been the dream of philosophers to have all of matter built up from one fundamental particle, so it is not altogether unsatisfactory that we have two in our theory. — P.A. M. Dirac

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Figure 1: Reed (2022), Fig. 52-39. “Notional sketch of a mass dipole formed by coupling of a positive mass and negative mass which interact to produce unidirectional thrust. The positive mass is attracted to the negative mass while the negative mass is repelled by the positive mass. Effective negative mass is created by negative index of refraction metamaterials which exhibit inverse Lorentz-Doppler effect. Metamaterials are composed of repeating elements of artificial structures which may be resonant or nonresonant. In the example illustrated, the repeating elements consist of transmission line LC filters: High pass left hand (LH) filter corresponds to positive mass; Low pass right hand (RH) filter corresponds to negative mass.” — L.J. Reed, *Quantum Wave Mechanics*, 4th ed., Ch. 52, p. 577.

In Plain English

Reed proposes that just as a magnetic dipole consists of a north and south pole, a “gravitational dipole” could consist of a region of positive gravitational mass coupled to a region of effective negative gravitational mass. In the Whittaker framework of GG Part II, this corresponds to a device that excites the longitudinal $(F + G)$ vacuum mode asymmetrically — putting more of the symmetric scalar mode on one side than the other. The asymmetry in the longitudinal mode creates a net gradient force in one direction, which is the propulsion mechanism.

In GG Part II’s language, Reed’s mass dipole is a device for generating an *asymmetric* ∇n_G — a spatial gradient in the gravitational GRIN index — without a large, symmetric gravitational mass as the source. The metamaterial structure emulates a $1/r$ GRIN profile over a finite volume by periodically varying the effective vacuum density at the sub-wavelength scale, using LC filter transmission lines whose impedance is engineered to match specific values of the local permittivity ε_G and permeability μ_G of the longitudinal Whittaker mode.

The physical mechanism is consistent with GG Part II’s Eq. (5.3): any localised gradient $\nabla n_G(\mathbf{r}) \neq 0$ produces a net gravitational force on all masses in its neighbourhood. Reed’s Figure 52-39 provides a concrete engineering schematic for realising such a gradient through metamaterial engineering of the vacuum’s longitudinal permittivity.

§ 12 The Mode-Energy Partition and the Derivation of G

12.1 Energy in Each Whittaker Mode

$$U_{\text{EM}} = m_e^{(0)} c_0^2 = \hbar \omega_C \quad (\text{transverse mode — all of the rest mass}) \quad (12.1)$$

$$U_G = \alpha_G \cdot m_e^{(0)} c_0^2 = \alpha_G \hbar \omega_C \approx 1.430 \times 10^{-78} \text{ J} \quad (\text{longitudinal mode}) \quad (12.2)$$

12.2 Newton’s G — A Structural Decomposition

Honest Status of this Result

The equation $G = \alpha_G \cdot \hbar c_0 / m_e^2$ is a genuine structural decomposition. It is not yet a full derivation, because α_G is still sourced from Cavendish measurement. The definition $\alpha_G \equiv G m_e^2 / (\hbar c_0)$ contains G itself. This will become a genuine derivation only once α_G can be derived from the three geometric primitives of Sankhya — identified in Part III as the main remaining open question.

Newton's G — Structural Decomposition (One Cavendish Input Required)

$$G = \alpha_G \cdot \frac{\hbar c_0}{m_e^{(0)2}} \quad \implies \quad G = 6.674300 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \quad (12.3)$$

where (to 7 significant figures):

$$\alpha_G = \frac{G m_e^2}{\hbar c_0} = 1.751810 \times 10^{-45} \quad (12.4)$$

$$\frac{\hbar c_0}{m_e^{(0)2}} = 3.811913 \times 10^{34} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \quad (12.5)$$

G is the inverse permittivity of the longitudinal vacuum mode. α_G is the single gravitational quantity that needs measuring from experiment.

§ 13 The Planck Scale as the GRIN Self-Confinement Threshold

Planck Mass as the GRIN Self-Trapping Threshold

$$m_{\text{Pl}} = \sqrt{\frac{\hbar c_0}{G}} = \frac{m_e^{(0)}}{\sqrt{\alpha_G}} = 2.176434 \times 10^{-8} \text{ kg} \quad (13.1)$$

This is the mass at which a particle's own gravitational field becomes strong enough to trap its own internal oscillation — the GRIN self-confinement threshold. Above it, no stable resonator is possible.

§ 14 The Hierarchy Gap in GRIN Language

The electromagnetic deflection length $\beta_{\text{EM}} = 5.635883 \times 10^{-15} \text{ m}$ is forty-three orders of magnitude larger than the gravitational deflection length $\beta_G = 1.352955 \times 10^{-57} \text{ m}$. This is the hierarchy of forces, stated in purely geometric terms. The framework gives no free parameters by which to adjust this ratio — it is entirely determined by $m_e/m_{\text{Pl}} = \sqrt{\alpha_G}$, an experimental fact.

§ 15 The Complete GRIN Dictionary: EM and Gravity Side by Side

Table 2: Complete GRIN Dictionary: Electromagnetic and Gravitational Sectors.

Concept	Electromagnetic Transverse ($F - G$)	—	Gravitational — Longi- tudinal ($F + G$)
Source “charge”	e		$m_e^{(0)}$
Vacuum permittivity	$\varepsilon_0 = 8.854188 \times 10^{-12} \text{ F/m}$		$\varepsilon_G = 1/(4\pi G) = 1.192297 \times 10^9 \text{ kg s}^2/\text{m}^3$
Coupling constant	$k_e = 1/(4\pi\varepsilon_0)$		$G = 1/(4\pi\varepsilon_G)$
Dim’less coupling	$\alpha_{\text{EM}} = k_e e^2/(\hbar c_0) = 7.297353 \times 10^{-3}$	=	$\alpha_G = G m_e^{(0)2}/(\hbar c_0) = 1.751810 \times 10^{-45}$
GRIN index	$n_{\text{EM}}(r) = 1 + 2\alpha_{\text{EM}} R_C^{(0)}/r$		$n_G(r) = 1 + 2\alpha_G R_C^{(0)}/r$
Deflection length	$\beta_{\text{EM}} = 2r_e = 5.635883 \times 10^{-15} \text{ m}$		$\beta_G = 2G m_e^{(0)}/c_0^2 = 1.352955 \times 10^{-57} \text{ m}$
Natural orbital scale	$a_0 = R_C^{(0)}/\alpha_{\text{EM}} = 5.291774 \times 10^{-11} \text{ m}$		$R_C^{(0)} = 3.861594 \times 10^{-13} \text{ m}$
Perihelion per orbit	$6\pi\alpha_{\text{EM}}^2 = 1.003765 \times 10^{-3} \text{ rad}$	$\times$	$6\pi\alpha_G = 3.302083 \times 10^{-44} \text{ rad}$
Mode energy	$U_{\text{EM}} = \hbar\omega_C = m_e^{(0)}c_0^2$		$U_G = \alpha_G \hbar\omega_C = 1.430 \times 10^{-78} \text{ J}$
Self-trapping thresh- old	$\alpha_{\text{EM}} = 1$ at Planck charge		$\alpha_G = 1$ at $m = m_{\text{Pl}}$
Sankhya Guna	Rajas — kinetic, charge mode		Tamas — potential, mass mode
White (2026) ana- logue	Transverse EM polarisation		Longitudinal acoustic $\omega = Dq^2$ dispersion

§ 16 Numerical Verification and Complete Scorecard

Table 3: Numerical predictions of the STAR–Russell–Whittaker–Dowdye framework vs observation. All theoretical values computed to 7 significant figures.

Prediction	Formula	Theory	Observed	Agmt
Mercury perihelion	$6\pi GM_\odot/[c_0^2 a(1 - e^2)]$	42.96000"/cy	42.980±0.001"/cy	0.05%
Solar light deflection	$4GM_\odot/(bc_0^2)$	1.750000"	1.75 ± 0.10"	< 0.1%
Gravitational redshift	$\Delta\nu/\nu = GM/(rc_0^2)$	2.120000×10^{-6}	2.12×10^{-6}	< 0.1%
Fine structure (GRIN)	$6\pi\alpha_{\text{EM}}^2$	1.003765 10 ⁻³ rad	× Sommerfeld 1916✓	< 0.01%
Bohr radius	$R_C^{(0)}/\alpha_{\text{EM}}$	5.291774 10 ⁻¹¹ m	× 5.291772 10 ⁻¹¹ m	× exact
Classical electron radius	$\alpha_{\text{EM}} R_C^{(0)}$	2.817941 10 ⁻¹⁵ m	× 2.817940 10 ⁻¹⁵ m	× exact
Newton's G	$\alpha_G \hbar c_0 / m_e^{(0)2}$	6.674300×10^{-11}	6.674300×10^{-11}	< 0.01%
Planck mass	$m_e^{(0)}/\sqrt{\alpha_G}$	2.176434 10 ⁻⁸ kg	× 2.176434 10 ⁻⁸ kg	× exact
$\alpha_G = (m_e/m_{\text{Pl}})^2$	$(9.109384 \times 10^{-31}/2.176434 \times 10^{-8})^2$	× 1.751810×10^{-45}	1.751800×10^{-45}	< 0.01%
White (2026) H levels	$\omega_n = \omega^*/n^2$	$ E_1 $ 13.59843 eV	= 13.59843 eV	exact
White (2026) Lyman- α	$f_{2 \rightarrow 1} = cR_H(1 - 1/4)$	2.466038 PHz	2.466038 PHz	exact
Electron ZBW perihelion	$6\pi\alpha_G$	3.302083 10 ⁻⁴⁴ rad	× — (prediction)	—

§ 17 The Complete Derivation Chain

[Physical axiom] Extinction-shift re-emission at local $c(r) = c_0/n(r)$ §1
 $\downarrow$
 [Cosmology] Russell cubic lattice; saddle surfaces at shared cube faces §2
 $\downarrow$
 [Optics] Biconvex GRIN lens at shared face; two converging cone wavefronts §3
 $\downarrow$
 [Formation] Sinc standing wave at focal point = STAR resonator §3.2
 $\downarrow$
 [Primitives] $\{\hbar, e, \omega_C\}$ wave-kinematic constants; c_0 = boundary condition
 $\downarrow$
 [White 2026] Acoustic vacuum confirms: longitudinal $1/c_s^2(r) = A + C/r$ §0b
 dispersion $\omega = Dq^2$ reproduces Rydberg ladder exactly
 $\downarrow$
 [Whittaker] Two polarisation sectors of scalar wave equation §11
 Transverse (F-G) $\rightarrow n_{EM}(r) = 1 + 2\alpha_{EM}R_C/r$ [Rajas / charge]
 Longitudinal (F+G) $\rightarrow n_G(r) = 1 + 2\alpha_G R_C/r$ [Tamas / mass]
 $\downarrow$
 [Fermat] GRIN ray eq. $\rightarrow$ Binet eq. $\rightarrow \delta\phi = 3\pi\beta/[a(1 - e^2)]$ §8
 $\downarrow$
 EM: $\beta = \beta_{EM}, a = a_0 \rightarrow \delta\phi_{EM} = 6\pi\alpha_{EM}^2$ (Sommerfeld) §9
 Grav: $\beta = \beta_G, a = R_C \rightarrow \delta\phi_{ZBW} = 6\pi\alpha_G$ (ZBW perihelion) §10
 Grav: $\beta = 2G^*M_{sun}/c_0^2 \rightarrow 42.96''/\text{cy}$ (Mercury perihelion, Part I)
 $\downarrow$
 [Mode energy] $G = \alpha_G \hbar c_0 / m_e^2$ (structural decomp.; α_G from Cavendish)
 [Planck] $m_{P1} = m_e / \sqrt{\alpha_G}$ (GRIN self-trapping threshold)
 $\downarrow$
 [One Cavendish input] $\alpha_G = 1.751810 \times 10^{-45}$ — all else is derived

§ 18 The Central Theorem

Central Theorem of the STAR–Russell–Whittaker–Dowdye Framework

The universe is a nested cubic lattice of extinction-shift GRIN wave-fields (Russell, 1953). At each shared face between adjacent cubes, the two neighbouring GRIN fields overlap to form a biconvex optical lens. Photon wavefronts from each cube converge through this lens as cones of light, focused to a common focal point. Their superposition at the focal point produces the Whittaker isotropic sinc standing wave — the STAR resonator — which is the physical realisation of a massive particle. The particle is not a primitive object; it is the stable fixed point of dual converging-cone optical focusing in a self-

sustaining GRIN cavity.

The resonator simultaneously emits into two Whittaker polarisation sectors of the vacuum: the antisymmetric transverse ($F - G$) sector (electromagnetism, coupling α_{EM}) and the symmetric longitudinal ($F + G$) sector (gravity, coupling α_G). Both sectors share the same sinc profile and the same Fermat-principle GRIN ray equation. The same Binet equation with coupling-specific deflection length β produces atomic fine structure, planetary perihelion, and the electron’s gravitational self-precession — all without special relativity, all without curved spacetime, all from Fermat’s principle in the extinction-shift GRIN vacuum.

Independent confirmation (March 9, 2026): White, Vera, Sylvester & Dudzinski have independently demonstrated in peer-reviewed Physical Review Research that the vacuum, treated as a longitudinal acoustic continuum with inverse sound speed $1/c_s^2(r) = A(\omega) + C(\omega)/r$ and quadratic dispersion $\omega = Dq^2$, generates the complete hydrogenic spectrum without quantum postulates. Their result is the acoustic dual of GG Part II’s GRIN optical description of the same electromagnetic sector.

Extension: The Verbelli Photon Bubble — The Photon Generates Its Own Acoustic Medium (Verbelli, 2024)

The White–Vera–Sylvester–Dudzinski result implicitly assumes that the acoustic vacuum density profile $\rho(r)$ is a fixed background sourced by the proton alone. Jason Verbelli’s independent analysis identifies a more fundamental picture: the longitudinal photon bubble is not propagating *through* a pre-existing acoustic medium — it is *constituting* that medium as it travels. The vacuum is quiescent between photon passages; the photon’s traversal is what makes it acoustically active. “Illumination in real time” is the precise content of this claim. Three complementary mathematical formulations establish it.

I. Self-Consistent Field: Stress-Energy Sourcing of the Acoustic Density

The photon field ψ_γ carries an electromagnetic energy density. In the extinction-shift GRIN vacuum, the relationship between energy density and refractive index is direct: the vacuum responds to the photon’s presence by locally modifying its acoustic density ρ . Writing the photon’s energy density as

$$T_\gamma^{00} = \frac{\hbar\omega}{V} |\psi_\gamma|^2, \quad (\text{V.1})$$

the self-sourced acoustic density field is

$$\rho(r, t) = \rho_0 + \frac{\lambda}{c_0^2} T_\gamma^{00}(r, t) = \rho_0 + \frac{\lambda\hbar\omega}{c_0^2 V} |\psi_\gamma(r, t)|^2, \quad (\text{V.2})$$

where λ is a vacuum-response coupling constant (dimensions of m^3) and ρ_0 is the unperturbed vacuum density. Since the refractive index satisfies $n^2(r, t) \approx 1 + \delta\rho/\rho_0$ for small perturbations, this gives

$$n^2(r, t) = 1 + \frac{\lambda\hbar\omega}{\rho_0 c_0^2 V} |\psi_\gamma|^2 + \frac{2\alpha_G R_C}{r}, \quad (\text{V.3})$$

where the final term is the pre-existing gravitational GRIN contribution from the proton. The photon now propagates in a medium whose refractive index depends on the photon's own amplitude. The self-consistent wave equation is

$$\left[\nabla^2 - \frac{n^2(r, |\psi_\gamma|^2)}{c_0^2} \partial_t^2 \right] \psi_\gamma = 0. \quad (\text{V.4})$$

This is an implicit nonlinear equation: ψ_γ appears on both sides. Its solution is not a free plane wave but a self-bound excitation — the Verbelli photon bubble.

II. Nonlinear Soliton: The Photon as a Self-Trapped Wave Packet

Equation (V.4) is rendered tractable in the slowly-varying envelope approximation. Writing $\psi_\gamma = \mathcal{A}(r, t) e^{i(k_0 x - \omega_0 t)}$, expanding $n^2 \approx n_0^2 + 2n_0 \chi_{\text{eff}} |\mathcal{A}|^2$ with $\chi_{\text{eff}} = \lambda\hbar\omega/(\rho_0 c_0^2 V n_0)$, and retaining leading-order dispersive and nonlinear terms yields the Gross–Pitaevskii / nonlinear Schrödinger equation for the photon envelope:

$$i \partial_t \mathcal{A} = -\frac{c_0}{2n_0 k_0} \nabla^2 \mathcal{A} - \frac{c_0 k_0 \chi_{\text{eff}}}{n_0} |\mathcal{A}|^2 \mathcal{A} + V_{\text{GRIN}}(r) \mathcal{A}. \quad (\text{V.5})$$

The term $-c_0 k_0 \chi_{\text{eff}}/n_0$ is a *defocusing nonlinearity in frequency space* but a *self-focusing nonlinearity in the amplitude equation*: it acts as an attractive potential proportional to $|\mathcal{A}|^2$ that prevents the wave packet from dispersing. The bright soliton solution — the exact photon bubble — is

$$\mathcal{A}(x, t) = \mathcal{A}_0 \text{sech}\left(\frac{x - v_g t}{\xi}\right) e^{i(k_s x - \omega_s t)}, \quad \xi = \sqrt{\frac{2c_0}{\chi_{\text{eff}} k_0 n_0 \mathcal{A}_0^2}}, \quad (\text{V.6})$$

propagating at the local group velocity $v_g = c_0/n_{\text{local}}$. The soliton width ξ is entirely determined by the photon's own amplitude $\mathcal{A}_0$ and the vacuum response coupling χ_{eff} : a higher-amplitude photon carves a narrower, more tightly self-bound bubble. The photon does not travel *through* the acoustic medium; it *is* the acoustic medium disturbance, propagating as a self-sustaining excitation in exact analogy with the STAR resonator at the Russell cubic lattice boundary.

III. Bootstrap Self-Consistency: The Dispersion Coefficient Inherits the GRIN Profile

In White et al.'s linear theory, the dispersion coefficient $D = \hbar/(2m_{\text{eff}})$ is a fixed background parameter. In the Verbelli extension, the effective acoustic mass is not fixed: it inherits the local refractive index via $m_{\text{eff}}(r) = m_0 n(r)$, so the dispersion coefficient

acquires a spatial dependence,

$$D(r) = \frac{\hbar}{2m_0 n(r)} = \frac{D_0}{n(r)}, \quad D_0 = \frac{\hbar}{2m_0}. \quad (\text{V.7})$$

Since $n(r)$ itself depends on the eigenfunction ψ_n via equation (V.3), the system is only self-consistent when ψ_n simultaneously (a) solves the eigenvalue problem with the given $D(r)$, and (b) generates, through its own probability density, precisely the $n(r)$ assumed in (a). The bootstrap condition is the coupled pair:

$$\delta n(r) = \chi_{\text{eff}} |\psi_n(r)|^2, \quad (\text{V.8a})$$

$$\left[-D(r) \nabla^2 + V_{\text{eff}}(r) \right] \psi_n = \omega_n \psi_n, \quad D(r) = \frac{D_0}{1 + \chi_{\text{eff}} |\psi_n|^2}. \quad (\text{V.8b})$$

In the weak-coupling limit $\chi_{\text{eff}} \rightarrow 0$, (V.8b) reduces to White et al.’s linear Schrödinger-analogue equation and the Rydberg ladder $\omega_n = \omega^*/n^2$ is recovered exactly. The first-order correction from the self-sourcing nonlinearity shifts each energy level by

$$\delta\omega_n = -\frac{\chi_{\text{eff}} D_0}{n_0} \int_0^\infty |\psi_n(r)|^4 \cdot 4\pi r^2 dr \equiv -\frac{\chi_{\text{eff}} D_0}{n_0} \Gamma_n, \quad (\text{V.9})$$

where $\Gamma_n = \int |\psi_n|^4 d^3r$ is the L^4 norm of the n th hydrogenic eigenfunction — a purely geometric quantity calculable analytically from the standard hydrogen wavefunctions. This is a **geometric nonlinear Lamb shift**: a correction to the Rydberg ladder that depends only on eigenfunction geometry and χ_{eff} , with no free parameters once λ is fixed by a single measurement.

Physical interpretation — “illumination in real time.” The phrase captures the following precise content. In the linear theory, the acoustic vacuum density profile $\rho(r)$ pre-exists the photon’s passage; the photon merely samples it. In the Verbelli extension, $\rho(r, t) - \rho_0$ vanishes identically in the absence of the photon: the medium only exists where and when the photon is present. The photon’s wavefront is therefore not a signal *in* the acoustic medium — it *is* the acoustic medium, advancing at $v_g = c_0/n_{\text{local}}$ as a self-bound soliton. There is no “dark” pre-polarised vacuum ahead of the photon to be traversed; the vacuum is activated, propagated through, and restored to quiescence in a single self-consistent event. This is the longitudinal-mode analogue of the STAR resonator’s self-sustaining sinc cavity: the photon bubble and the particle resonator are the propagating and standing-wave limits of the same self-sourced acoustic excitation.

§ 19 Conclusions

This paper has shown that the reconciliation of electromagnetism and gravity does not require a new fundamental principle. It requires recognising the correct physical foundation: the extinction-shift re-emission mechanism, which makes the speed of light a locally-varying field $c(r) = c_0/n(r)$ rather than a universal constant. On this foundation, the following five conclusions follow.

First, particles are not primitive. They are the focal points of biconvex GRIN lenses formed at the boundaries of Russell’s cubic wave-fields — the standing-wave result of two counter-propagating cone wavefronts converging through the same lens from opposite sides. Russell described this in 1953 from cosmological first principles; the extinction-shift GRIN framework provides the optical mechanism that explains why the geometry is stable and self-reinforcing.

Second, electromagnetism and gravity are not two forces but two polarisation modes. They are the antisymmetric and symmetric Whittaker sectors of the same scalar wave equation in the extinction-shift vacuum. Newton’s G is the inverse permittivity of the longitudinal sector. The decomposition $G = \alpha_G \hbar c_0 / m_e^{(0)2}$ separates G into a purely electromagnetic scale $G_e = \hbar c_0 / m_e^2$ and a dimensionless coupling α_G . The programme’s next goal is deriving α_G from the three geometric primitives of Sankhya.

Third, Sommerfeld’s fine structure is not a relativistic result. It is an optical GRIN ray-bending result: the electron’s orbit in the Coulomb GRIN field fails to close by exactly $6\pi\alpha_{\text{EM}}^2 = 1.003765 \times 10^{-3}$ rad per circuit — derivable from Fermat’s principle alone.

Fourth, the Sankhya framework is structurally vindicated at five independent points. The cubic substratum, the self-sustaining resonator (Suthra 31), the dual polarisation of mass and charge (Suthra 52), the identification of imbalance as the cause of all observable physics (Suthra 45), and the interpretation of ω_C as the Sattva invariant threading both modes — all five are claims Sankhya makes axiomatically and this framework derives geometrically.

Fifth, the White–Vera–Sylvester–Dudzinski (2026) peer-reviewed result provides independent, published confirmation of the acoustic-vacuum sector. Their exact isospectrality result — quantisation emergent from quadratic dispersion $\omega = Dq^2$ in a Coulombic acoustic continuum, with no quantum postulates — is the acoustic dual of the GG Part II extinction-shift GRIN picture. GG Part II was formulated independently five days earlier and contains White et al.’s result as a special case, extending it to the gravitational sector, particle formation, and the cosmological Russell cubic lattice.

The outstanding open question is why the Russell lattice produces STAR resonators at the ZBW frequency $\omega_C = 7.763437 \times 10^{20}$ rad/s rather than at the Planck frequency. This is the hierarchy problem restated in geometric terms, and it is the invitation for Part III of this work.

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18. Part I of this series: “The Perihelion of Mercury from the STAR Sinc Field” (GRIN ray equation, 42.96"/cy).
19. Part III of this series: “The Genesis of Mass — Lorentz-Variant Derivation of α , m_e , α_G , and G ” (local mass $m(r) = m_e^{(0)} n(r)^2$).

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STAR Lattice Framework · Series Paper II · Enhanced Edition with White et al. (2026) Integration